Holo Lab User Guide

Holo Lab is a holographic workbench in your browser. You control 3D holograms with your bare hands through a webcam. You can grab them, throw them, twist them, resize them, take them apart and look inside. This guide covers setup, every gesture and key, the Holo Shooter game, voice requests, and fixes for common problems.

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1. What you need

- **Browser:** desktop **Chrome** or **Edge** works best. Other browsers may load, but hand tracking and speech can be slower or may not work.
- **Webcam:** any built-in or USB camera. An iPhone used as a Continuity Camera also works, and it can run at 60 FPS.
- **Space and light:** sit about an arm's length from the camera in an evenly lit room. Your hands should be fully in the frame. A plain background helps.
- **Microphone (optional):** needed only for voice requests.
- **No camera?** You can use a mouse instead. See section 10.

2. Starting Holo Lab

Online

1. Open **https://malikSumit3000.github.io/holo-lab/**.
2. Click **INITIALIZE** on the boot screen.

3. When the browser asks, **allow camera access**.
4. The boot log shows each system coming online: hand tracking, camera, and the object library. A **Planet** and an **Arc Reactor** then appear.
5. When the hint says *"PINCH AN OBJECT TO GRAB IT"*, you're ready.

If the camera is blocked or the hand model fails to load, Holo Lab switches to **mouse mode** and tells you so in the boot log.

Locally (Mac version)

See section 13. The local version adds AI-designed objects and lets you use your own 3D models.

3. The screen

Area	Where	What it shows
Status panel	top left	Tracking type (hands or mouse), hands seen, object count, current mode (MANIPULATE, LIGHT PEN, DRILL DEPTH n, HOLO SHOOTER), render FPS and hand-tracking FPS
Gestures panel	right	A quick gesture and key list. Press `H` to hide or show it
Hand cursors	follow your hands	A ring for each hand. It fills up while you hold a fist
Breadcrumbs	top centre	Your position inside an object, for example `V4 ENGINE › PISTON ASSEMBLY`. Pinch a crumb to jump back to it
Info card	right	The name, purpose, material and a fun fact for the part you're focused on or hovering over. Library models also show their licence
Dock	bottom	Buttons: PLANET, REACTOR, ENGINE, DNA, HYPERCUBE, VOICE, LIGHT PEN, GAME, CLEAR
Command bar	bottom	MIC button and a text box for requests, plus status messages and suggestion chips

You can **pinch any dock button, breadcrumb or chip with your hand** to press it, or click it with the mouse.

Up to **10 objects** can be on screen at once. To make room, disintegrate some or press `X` to clear them all.

4. Hand gestures

A **pinch** means touching your thumb tip to your index fingertip. Keep your index finger reaching out a little: a tightly curled fist doesn't count as a pinch.

Gesture	What it does
Pinch an object	Grab it. Move your hand to drag it
While holding: **push / pull** your hand toward or away from the camera	Move the object nearer or further away
While holding: **twist** your hand	Rotate the object
Flick your hand and let go	Throw the object. It bounces off the screen edges
Open palm swipe across an object	Spin it
Two-hand pinch on one object, then **spread or squeeze** your hands	Resize it. On objects with parts, spreading **explodes** it into its parts
Fist, held for 1 second over an object	Disintegrate (delete) it. The cursor ring fills while you hold
Pinch a dock button	Press it

Tips:

- **Grabbing is forgiving.** The grab zone is a little larger than the object, and the nearest object wins.
- **Grabs are "sticky."** A brief tracking flicker won't drop the object. It's released only after your fingers stay open for a moment (80 ms by default).
- Small, unintended depth changes and twists are ignored while you hold an object, so it stays still when your hand does.

5. Taking objects apart (drill-down)

Every object is built as a tree of parts. The **V4 Engine** is the deepest, three to four levels down. Library models with real part structures, such as the Milk Truck, can be taken apart as well. Single-piece models are marked *solid*. You can grab, twist and resize them, but they don't explode.

1. **Explode:** two-hand pinch the object and spread your hands. Parts fly apart and labels appear.
2. **Pull out a part:** pinch one part of the exploded object and pull it away.
    - Let go **near its slot** and it snaps back into place.
    - Let go **far from its slot** to **drill in**. The part flies to the centre, enlarges and becomes the focus, and everything else dims to a ghost.
3. **Explode again:** spread the focused part with two hands to see its sub-parts. Repeat as deep as the object goes.
4. **Go back up:** press `B` or `Esc`, pinch ◀ **BACK**, or pinch any breadcrumb. The part flies back into its slot.

While you're drilled in, the dimmed objects in the background can't be grabbed, so you can't move them by accident. The info card describes whatever you're focused on or hovering over.

Try this: spawn the engine (`3`), explode it, pull out a *Piston assembly*, drill in, explode it, pull out the *Connecting rod*, drill in again, then press `B` twice to return to the full engine.

6. The light pen

The light pen lets you sculpt glowing strokes in 3D space.

1. Press `P` or pinch **LIGHT PEN** in the dock. The mode changes to LIGHT PEN.
2. **Pinch in empty space** and move your hand to draw. Push or pull your hand to draw in depth.
3. Press `P` again to switch the pen off. Press `X` or **CLEAR** to remove everything.

7. Asking for objects by voice or text

Speaking

- **Hold** `SPACE`, say what you want, then **release** it. For example: *"show me a truck"*, *"an engine"*, *"spawn a lantern"*.
- You can also pinch **MIC** (or the **VOICE** dock button) to start, then pinch it again to stop.
- A level meter shows it's listening. The text it heard appears in the command bar so you can check it.
- The first time you speak, the **Whisper** speech model downloads once (about 77 MB) and is cached. After that, speech recognition runs entirely in your browser and works offline.

Typing

- Press `/` (or click the text box), type a request, and press **Enter**. Press `Esc` to leave the box.

What happens next

- **Clear match:** the object appears straight away.
- **Close match:** you get *"Did you mean…"* chips. Pinch or click one.
- **No match:**
  - **Online:** you'll see suggestions from the library.
  - **Mac version with Ollama:** your local AI designs the object from simple shapes, which takes roughly 15–45 seconds. The design is a real part tree that you can explode and drill into. Designs are cached, so asking again is instant.

What's in the library

Built-in (always available)	Online models (Khronos glTF samples)
Planet (`1`), Arc Reactor (`2`), V4 Engine (`3`), DNA Helix (`4`), Tesseract/Hypercube (`5`)	Antique Camera, Avocado, Boom Box, Lantern, Milk Truck, Toy Car, Water Bottle

Each object also answers to related words. "Earth", "globe", "motor", "car", "radio", "lamp" and "bottle" all work.

8. Holo Shooter (game mode)

Press `G` or pinch **GAME**. Your lab objects are hidden and kept safe until you return.

Controls

Action	Hand	Mouse / keys
Aim	Make a finger gun: index finger out, other fingers curled. The crosshair follows where your finger points	Move the mouse
Fire	**Drop your thumb** onto your hand like a hammer, or **flick** your finger up like recoil. Lift your thumb again before the next shot	Left click
Shield	**Open palm** with your other hand. A hex shield blocks incoming shots in its area	Right button
Reload	Make a **fist** with your other hand (the magazine holds 8 rounds)	`R`
Exit	Pinch or shoot **✕ EXIT**	`Esc` or `G`

Gameplay

- Holographic drones attack in **waves** at different depths. Some hover, some strafe and some dive. They charge up, then fire slow plasma shots at you.
- **Red** drones are basic (100 points), **orange** armoured drones take 3 hits (250 points), and every 5th wave brings a **pink boss** (2500 points).
- The HUD shows **score, wave, combo multiplier, accuracy and best score**, plus your **HULL** (health) and ammo.
- When your hull runs out, the **GAME OVER** screen shows your results. Shoot or click **RESTART** to play again. Your best score is saved in this browser.
- In game mode, `D` shows aiming debug info, and `M`, `F` and `T` still work.

If aiming feels too twitchy or too sluggish, open **Tracking tuning** (`T`) and adjust the *Shooter* sliders (section 11).

9. 3D window (head tracking)

Press `W` to turn the screen into a **window into a 3D room**. When you move your head, the view shifts as if you were looking through glass. The first time you turn it on, a small face-tracking model (about 4 MB) loads. It works in both the lab and the game. Press `W` again to turn it off. It needs the camera.

10. Using a mouse instead of a camera

Mouse	Lab	Game
Move	Cursor	Aim
Left click / drag	Pinch: grab, drag, throw (release while moving), press buttons, draw with the pen	Fire
Right drag	Open palm: spin objects	Shield
Wheel	While dragging: move in depth. Over an object: resize it, or explode/assemble it if it has parts	—
`Delete` key	Delete the object under the cursor	—

11. Tracking tuning

Press `T` to open the tuning panel. Changes take effect immediately and are **saved in this browser**. **RESET DEFAULTS** restores the factory settings.

Slider	Raise it if…	Lower it if…
Smoothing (steadiness)	—	The cursor jitters when your hand is still
Responsiveness (less lag)	Objects lag behind fast moves	The cursor gets jittery
Pinch sensitivity	Pinches aren't detected	Grabs happen by accident
Sticky grab (ms)	Objects drop while you hold them	Objects are slow to release
Depth sensitivity	Push/pull moves objects too little	Objects shoot in and out
Twist sensitivity	Rotation feels sluggish	Rotation feels twitchy
Grab zone size	Objects are hard to grab	You grab the wrong object
Shooter: aim assist	You want more help hitting targets	The crosshair feels pulled around
Shooter: aim reach	You must move your hand far to cross the screen	Aim is too twitchy
Shooter: finger-direction aiming	You want aim to follow where your finger points	You'd rather aim by hand position
Shooter: thumb trigger sensitivity	Thumb shots don't register	Shots fire by accident
Shooter: flick-to-fire speed	Shots fire by accident	Flicks don't register

The panel's own rule of thumb: *if it's slow and steady, lower smoothing; if it lags on fast moves, raise responsiveness.*

12. Keyboard reference

Lab

Key	Action
`1` – `5`	Spawn Planet, Arc Reactor, V4 Engine, DNA Helix, Tesseract
`SPACE` (hold)	Speak a request
`/`	Type a request
`B` / `Esc`	Go back up one drill level
`P`	Toggle light pen
`X`	Clear all objects
`G`	Start Holo Shooter
`W`	Toggle 3D window (head tracking)
`H`	Hide or show the gestures panel
`T`	Tracking tuning panel
`V`	Toggle the camera background video
`C`	Switch to the next camera
`M`	Mute or unmute sound
`F`	Toggle fullscreen
`Delete`	Delete the object under the mouse

Game

Key	Action
`R`	Reload
`Esc` / `G`	Back to the lab
`D`	Aiming debug overlay
`M` / `F` / `T` / `W`	Mute / fullscreen / tuning / 3D window

13. Running the Mac (local) version

The local version does everything the online version does, and adds the following:

- **AI designs:** objects that aren't in the library are designed by a local AI model.
- **Your own model library:** download the curated set of models, or add your own `.glb` files.

Setup

```
git clone https://github.com/malikSumit3000/holo-lab.git
cd holo-lab
python3 tools/fetch_library.py   # optional: download the 3D model library (~64 MB, asks first)
python3 run.py                   # starts a local server and opens http://localhost:8765
```

`run.py` uses only the Python standard library. It serves the app on `localhost`, which browsers allow to use the camera. It also opens Chrome if it can find it.

Option	Meaning
`--port 8765`	Preferred port. If it's busy, the next free port is used
`--no-browser`	Don't open a browser automatically
`HOLO_MODEL=qwen3:14b python3 run.py`	Use a specific Ollama model

Stop the server with `Ctrl+C`, or with the Stop button in PyCharm.

AI designs with Ollama

1. Install Ollama.
2. Pull a model: `ollama pull qwen3`.
3. Start `run.py`. It starts Ollama for you if it isn't already running, and it chooses from qwen3, qwen2.5, llama3, gemma or mistral, in that order.

Designs from an 8B model are simple and occasionally odd. A larger model, such as `qwen3:14b`, gives better results if your Mac has enough memory.

Adding your own 3D models

1. Put any `.glb` file into `library/models/`.
2. Run `python3 tools/fetch_library.py --scan`.
3. The file name becomes the model's voice name. For example, `jet_engine.glb` answers to *"show me a jet engine"*.

To download only some of the curated models, use `--only toy_car,lantern`. To skip the confirmation prompt, use `--yes`.

14. Troubleshooting

Problem	Fix
Boot log says **"No camera access - using mouse instead"**	Allow the camera in the browser's site settings and reload. On macOS, also allow Chrome under **System Settings → Privacy & Security → Camera**. Close other apps that are using the camera
Wrong camera in use	Press `C` to switch to the next camera
Hands not detected	Improve the lighting, move back so your whole hand is visible, and keep your palm facing the camera. Check that **HAND TRACKING** in the status panel shows an FPS value
Cursor is **jittery** or **laggy**	Adjust **Smoothing** and **Responsiveness** in the tuning panel (`T`)
Objects **drop** while held	Raise **Sticky grab** and/or lower **Pinch sensitivity**
Accidental grabs	Lower **Pinch sensitivity** or **Grab zone size**
"MAX OBJECTS"	You've reached 10 objects. Disintegrate some with a fist hold, or press `X`
"Microphone blocked"	Allow the microphone in the browser. On macOS, also allow Chrome under **System Settings → Privacy & Security → Microphone**
"Didn't catch that"	Hold `SPACE` a little longer and speak clearly. The first request waits for the speech model to download
Online: *"isn't in the online library"*	Pick one of the suggested chips, or use the Mac version with Ollama for AI designs
Mac version: AI generation fails	Make sure Ollama is installed and has a model (`ollama pull qwen3`), then restart `run.py`
Low FPS	Use fewer objects, close other heavy tabs, press `V` to hide the camera background, and use Chrome or Edge with hardware acceleration enabled
Can't grab a dimmed object	You're drilled into a part. Press `B` / `Esc` to go back up first
"USE BACK TO LEAVE THIS PART FIRST"	The focused part can't be deleted. Go back up a level first

For developers: open the app with `?test` at the end of the URL (for example `http://localhost:8765/?test`) to run the built-in gesture regression tests with synthetic hands.

15. Privacy and licences

- **Video and audio stay on your computer.** Hand tracking, face tracking and speech recognition all run inside your browser. The camera image and your voice are never uploaded.
- Model files (hand/face tracking, Whisper, and the online 3D models) are downloaded once and cached by the browser.
- Your tuning settings and best game score are stored only in this browser's local storage.
- The Mac version's AI requests go only to Ollama on your own machine.

- **3D model licences:** the online models come from the Khronos glTF Sample Assets. Six are CC0. The Cesium Milk Truck is CC-BY-4.0 (© Cesium). Each model's licence is shown in its info card and listed in `web-library.json`. Downloaded models are recorded by `tools/fetch_library.py`, and non-commercial licences are flagged in the info card.

Built with MediaPipe Tasks, Three.js, transformers.js (Whisper) and Ollama.